Advanced Statistical Methods Summary Notes

Contents

1 Basic Setup

The simplest setup for all subsequent discussions is that we have a *population* with some associated (unknown) *parameters*.

Following are some parameters associated with a population that could be of our interest.

1. Mean (μ)
2. Variance (σ^2)
3. *Binomial parameter* p

The last item in the list deserves a small remark. Suppose we can bifurcate the population in two categories, where one category perhaps favors some policy and the other one does not. Then the proportion that favors the policy can be thought of as a binomial random variable.

But sometimes, we are interested in comparing parameters of two different populations. That is, we have two independent populations with associated means μ_1 and μ_2 and variances σ_1^2 and σ_2^2. Perhaps there are also two underlying binomial parameters p_1 and p_2 with each population. Then following could be some parameters of interest.

1. $\mu_1 - \mu_2$
2. $\sigma_1^2 - \sigma_2^2$
3. $p_1 - p_2$

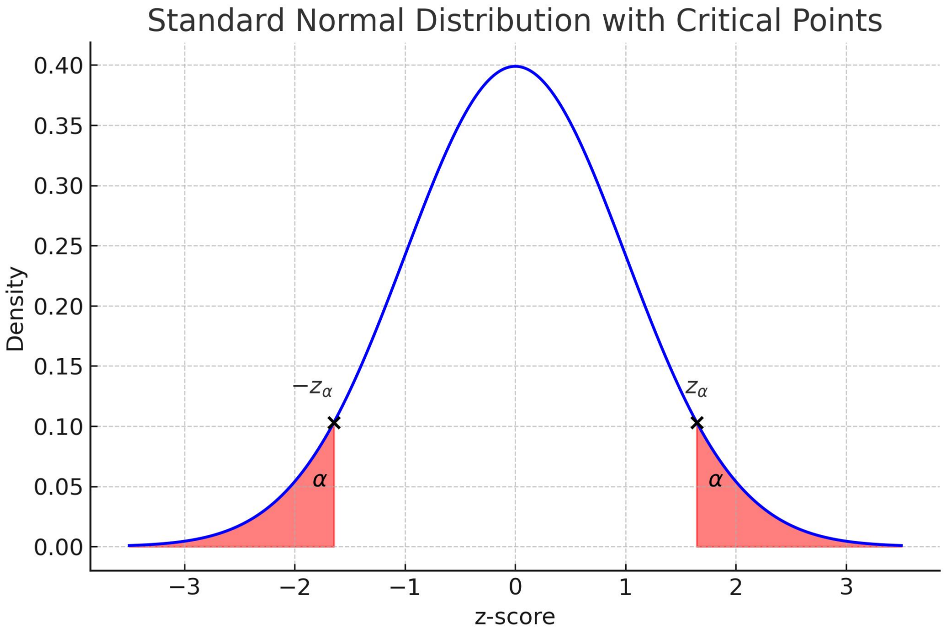
Our objective is to get as better of a sense as possible for the values of these parameters.

When we are concerned with just one population, we start off by collecting a (random) *sample* of size n. Let this be denoted by $Y_1, Y_2, \cdots, Y_n$ and more formally, each Y_i is a random variable with some underlying (generally unknown) distribution. For simplicity we always assume that all these Y_i's are iid random variables.

Now if we are concerened with two populations, then we collect two independent random sample from each population of sizes n_1 and n_2. We can denote the sample from the first population as, $Y_{11}, Y_{12}, \cdots, Y_{1n_1}$ and the sample from the second population as, $Y_{21}, Y_{22}, \cdots, Y_{2n_2}$.

1.1 Few Common Notations

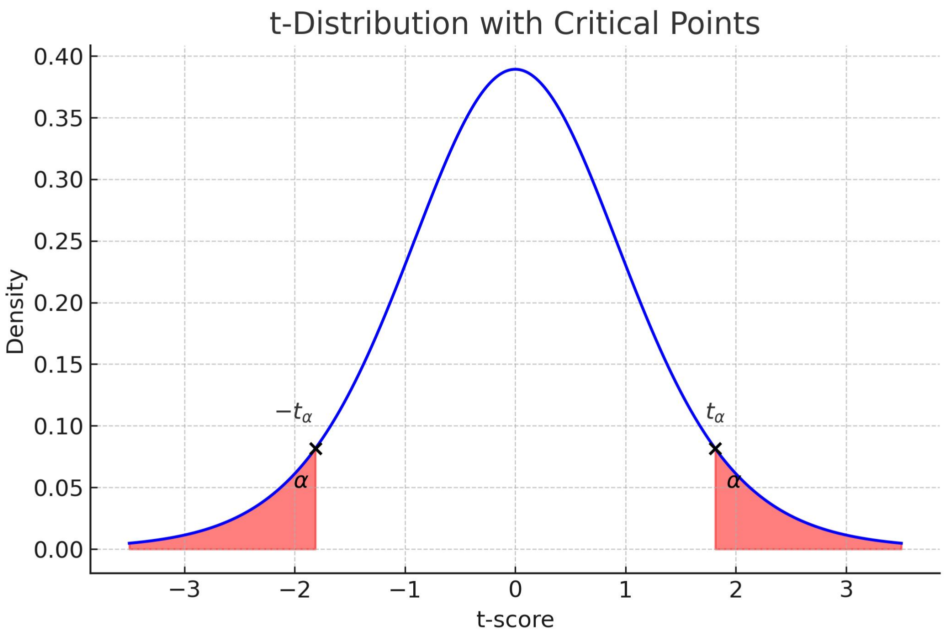
1.1.1 Normal Distribution



The points $-z_\alpha$ and z_α can be computed as,

$$-z_\alpha = qnorm(\alpha)$$
$$z_\alpha = qnorm(1-\alpha)$$

1.1.2 t Distribution

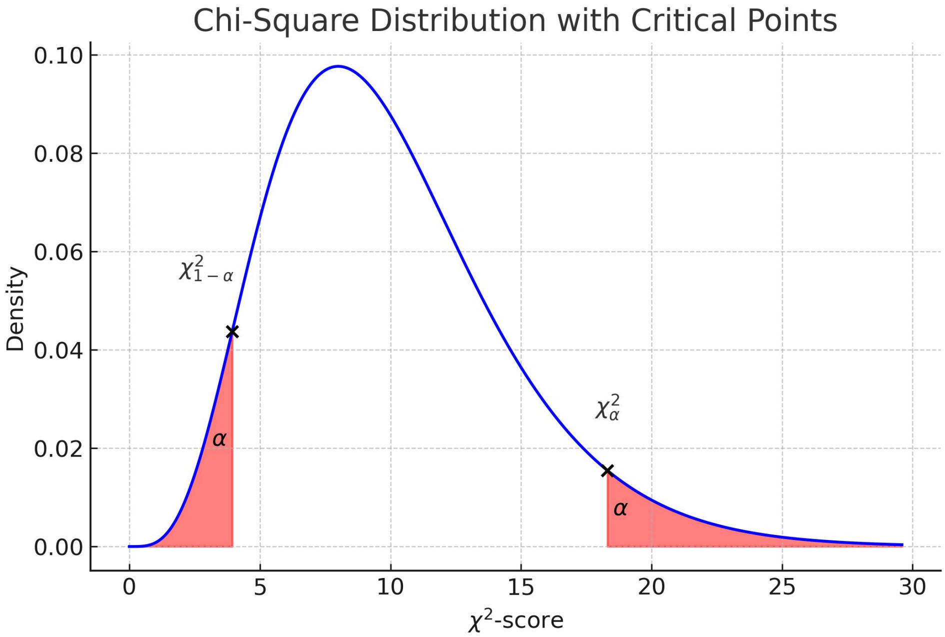


The points $-t_\alpha$ and t_α can be computed as,

$$-t_\alpha = qt(\alpha, \nu)$$
$$t_\alpha = qt(1-\alpha, \nu)$$

where, ν denotes the associated degree of freedom of the t-distributed random variable.

1.1.3 Chi-square Distribution

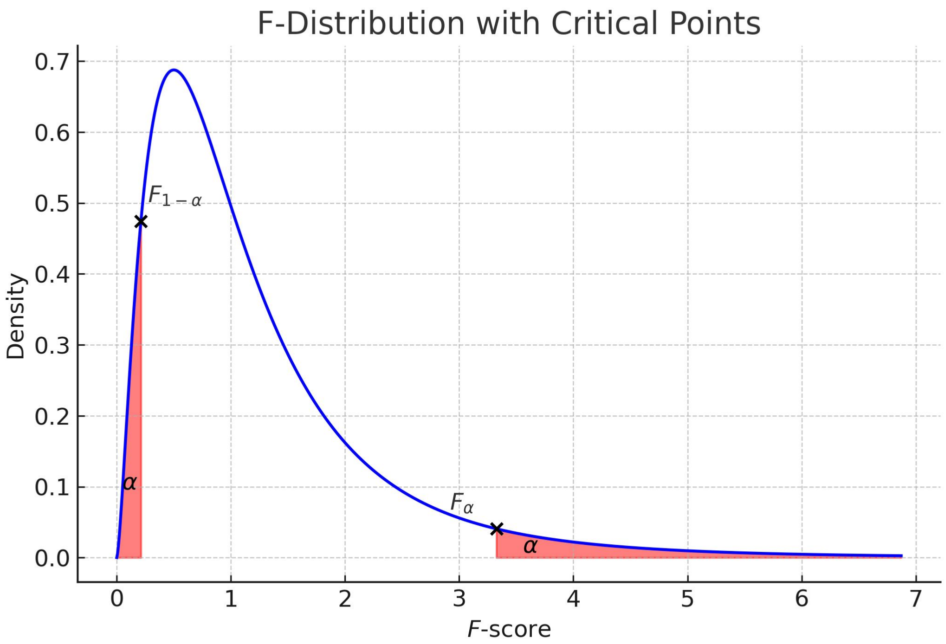


The points $\chi^2_{1-\alpha}$ and χ^2_α can be computed as,

$$\chi^2_{1-\alpha} = qchisq(\alpha, \nu)$$
$$\chi^2_\alpha = qchisq(1-\alpha, \nu)$$

where, ν denotes the associated degree of freedom of the χ^2-distributed random variable.

1.1.4 F Distribution



The points $F_{1-\alpha}$ and F_α can be computed as,

$$F_{1-\alpha} = qf(\alpha, \nu_1, \nu_2)$$
$$F_\alpha = qf(1-\alpha, \nu_1, \nu_2)$$

where, ν_1 denotes the associated numerator degree of freedom and ν_2 denotes the associated denominator degree of freedom of the F-distributed random variable.

2 Point Estimators

2.1 Point Estimator for Mean

To estimate population mean μ, we use,

$$\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

We have,

$$\mathbb{E}\left[\overline{Y}\right] = \mu$$

and,

$$Var(\overline{Y}) = \sigma^2/n$$

2.2 Point Estimator for Variance

To estimate population variance (σ^2), we use,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(Y_i - \overline{Y} \right)^2$$

We have,

$$\mathbb{E}\left[S^2\right] = \sigma^2$$

2.3 Point Estimator for Binomial Parameter

To estimate binomial parameter p associated with a population, we collect a random sample $Y_1, Y_2, \cdots, Y_n$ where each Y_i is a Bernoulli random variable, and then we can use,

$$\hat{p} = \overline{Y} = \frac{\sum_{i=1}^{n} Y_i}{n}$$

We have,

$$\mathbb{E}\left[\hat{p}\right] = p$$

and,

$$Var(\hat{p}) = \frac{p(1-p)}{n}$$

2.4 Point Estimator for Difference of Means

To estimate $\mu_1 - \mu_2$, we use,

$$\overline{Y_1} - \overline{Y_2} = \left(\frac{1}{n_1} \sum_{i=1}^{n_1} Y_{1i} \right) - \left(\frac{1}{n_2} \sum_{j=1}^{n_2} Y_{2j} \right)$$

We have,

$$\mathbb{E}\left[\overline{Y_1} - \overline{Y_2}\right] = \mu_1 - \mu_2$$

and,

$$Var(\overline{Y_1} - \overline{Y_2}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

2.5 Point Estimator for Difference of Variances

To estimate $\sigma_1^2 - \sigma_2^2$, we can use,

$$S_1^2 - S_2^2 = \left(\frac{1}{n_1 - 1} \sum_{i=1}^{n_1} \left(Y_{1i} - \overline{Y_1} \right)^2 \right) - \left(\frac{1}{n_2 - 1} \sum_{j=1}^{n_2} \left(Y_{1j} - \overline{Y_2} \right)^2 \right)$$

We have,

$$\mathbb{E}\left[S_1^2 - S_2^2\right] = \sigma_1^2 - \sigma_2^2$$

2.6 Point Estimator for Difference of Binomial Parameters

To estimate $p_1 - p_2$, we again collect two independent random samples from each population, $Y_{11}, Y_{12}, \cdots, Y_{1n_1}$, from the first population and $Y_{21}, Y_{22}, \cdots, Y_{2n_2}$, from the second population, where each Y_{ij} is again a Bernoulli random variable and then we use,

$$\hat{p_1} - \hat{p_2} = \left(\frac{\sum_{i=1}^{n_1} Y_{1i}}{n_1}\right) - \left(\frac{\sum_{j=1}^{n_2} Y_{2j}}{n_2}\right)$$

We have,

$$\mathbb{E}\left[\hat{p_1} - \hat{p_2}\right] = p_1 - p_2$$

and,

$$Var(\hat{p_1} - \hat{p_2}) = \left(\frac{p_1(1-p_1)}{n_1}\right) + \left(\frac{p_2(1-p_2)}{n_2}\right)$$

3 Interval Estimation

3.1 Interval Estimation for Estimators following Normal Distribution

Consider following parameters,

1. Mean μ
2. Binomial Parameter p
3. $\mu_1 - \mu_2$
4. $p_1 - p_2$

Let's denote the above parameters in general as θ and their corresponding point estimators as $\hat{\theta}$.

If the sample size, that is $n \geq 30$, then we have following useful results.

$\hat{\theta}_L$, such that,

$$\Pr\left(\hat{\theta}_L \leq \theta\right) = 1 - \alpha$$

is given as,

$$\hat{\theta}_L = \hat{\theta} - z_\alpha \cdot SD(\hat{\theta})$$

Similary $\hat{\theta}_U$, such that,

$$\Pr\left(\theta \le \hat{\theta}_U\right) = 1 - \alpha$$

is given as,

$$\hat{\theta}_U = \hat{\theta} + z_\alpha \cdot SD(\hat{\theta})$$

.

Finally, $\hat{\theta}_L$ and $\hat{\theta}_U$, such that,

$$\Pr\left(\hat{\theta}_L \le \theta \le \hat{\theta}_U\right) = 1 - \alpha$$

is given as,

$$\begin{aligned} \hat{\theta}_L &= \hat{\theta} - z_{\alpha/2} \cdot SD(\hat{\theta}) \\ \hat{\theta}_U &= \hat{\theta} + z_{\alpha/2} \cdot SD(\hat{\theta}) \end{aligned}$$

If the sample size, $n < 30$, then we have following useful results.

a and b such that,

$$\Pr\left(a \le \mu \le b\right) = 1 - \alpha$$

is given as,

$$\begin{aligned} a &= \overline{Y} - t_{\alpha/2}\left(\frac{S}{\sqrt{n}}\right) \\ b &= \overline{Y} + t_{\alpha/2}\left(\frac{S}{\sqrt{n}}\right) \end{aligned}$$

where $\nu = n - 1$.

Similarly, a and b such that,

$$\Pr\left(a \le \mu_1 - \mu_2 \le b\right) = 1 - \alpha$$

is given as,

$$\begin{aligned} a &= \left(\overline{Y_1} - \overline{Y_2}\right) - t_{\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ b &= \left(\overline{Y_1} - \overline{Y_2}\right) + t_{\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \end{aligned}$$

where $\nu = n_1 + n_2 - 2$ and $S_p = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$

3.2 Interval Estimation for Variance

Suppose we want to find a confidence interval around σ^2 with confidence coefficient $1-\alpha$, i.e., we want to find some a and b such that,

$$\Pr\left(a \leq \sigma^2 \leq b\right) = 1 - \alpha$$

Then, we have,

$$a = \frac{(n-1)S^2}{\chi^2_{\alpha/2}}$$
$$b = \frac{(n-1)S^2}{\chi^2_{1-(\alpha/2)}}$$

where $\nu = n - 1$.

4 Hypothesis Testing

4.1 Hypothesis Testing Concerning Estimators that follow Normal Distribution

Once again consider the following parameters,

1. Mean μ
2. Binomial Parameter p
3. $\mu_1 - \mu_2$
4. $p_1 - p_2$

Let's once again denote these parameters as θ and let $\hat{\theta}$ be their corresponding estimators.

To test H_0 versus H_a with *significance level* α, we use a *test statistic* and define a *rejection region* such that, if the test statistic lies in the rejection region then the *null hypothesis*, H_0, is rejected otherwise we fail to reject H_0.

For any test associated with any one of the four parameters stated above, Suppose $H_0 := \theta = \theta_0$. Then we use the following test statistic,

$$Z = \frac{\hat{\theta} - \theta_0}{SD(\hat{\theta})}$$

Now if $H_a := \theta < \theta_0$. Then the rejection region is defined as,

$$Z < -z_\alpha$$

If instead, $H_a := \theta > \theta_0$, then the rejection region is defined as,

$$Z > z_\alpha$$

Finally, if $H_a := \theta \neq \theta_0$, then the rejection region is defined as,

$$Z < -z_{\alpha/2} \text{ or } Z > z_{\alpha/2}$$

But then the above test statistic only works well if the sample size, $n \geq 30$.

Now if the sample size is *small*, then, we have following test concerned with the population mean μ.

Suppose $H_0 := \mu = \mu_0$, then using the test statistic,

$$T = \frac{\overline{Y} - \mu_0}{S/\sqrt{n}}$$

where $\nu = n - 1$
If we have $H_a := \mu < \mu_0$, then we define the rejection region as,

$$T < -t_\alpha$$

If instead, we have $H_a := \mu > \mu_0$, then we define the rejection region as,

$$T > t_\alpha$$

Finally, if $H_a := \mu \neq \mu_0$, then the rejection region is defined as,

$$T < -t_{\alpha/2} \text{ or } T > t_{\alpha/2}$$

An important assumption in this test is that the underlying distribution of the random samples is an approximately normal distribution.

Suppose we want to compare means of two different populations and the sample size is *small*.

If $H_0 := \mu_1 - \mu_2 = D_0$, then using the test statistic,

$$T = \frac{\overline{Y_1} - \overline{Y_2} - D_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where $\nu = n_1 + n_2 - 2$ and $S_p = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$

For $H_a := \mu_1 - \mu_2 < D_0$, we define the rejection region as,

$$T < -t_\alpha$$

If $H_a := \mu_1 - \mu_2 > D_0$, we define the rejection region as,

$$T > t_\alpha$$

If $H_a := \mu_1 - \mu_2 \neq D_0$, we define rejection region as,

$$T < -t_{\alpha/2} \text{ or } T > t_{\alpha/2}$$

An important assumption in this test is that the underlying distribution of the random samples is a normal distribution and further more, variance of both populations is *same*.

4.2 Hypothesis Tests Concerning Variances

Suppose $H_0 := \sigma^2 = \sigma_0^2$. Then considering the test statistic,

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

If $H_a := \sigma^2 < \sigma_0^2$, then we can define the rejection region as,

$$\chi^2 < \chi^2_{1-\alpha}$$

If $H_a := \sigma^2 > \sigma_0^2$, then we define the rejection region as,

$$\chi^2 > \chi^2_\alpha$$

And, if $H_a := \sigma^2 \neq \sigma_0^2$, then we define the rejection region as,

$$\chi^2 < \chi^2_{1-\alpha/2} \text{ or } \chi^2 > \chi^2_{\alpha/2}$$

Again, an important assumption is that underlying population distribution is a normal distribution.

Suppose we have two populations with variances σ_1^2 and σ_2^2, respectively and suppose $H_0 := \sigma_1^2 = \sigma_2^2$. Then considering the test statistic,

$$F = \frac{S_1^2}{S_2^2}$$

we have that it follows F-distribution with $(n_1 - 1)$ numerator *df* and $(n_2 - 1)$ denominator *df*.
If $H_a := \sigma_1^2 < \sigma_2^2$, then we define the rejection region as,

$$F > F_{1-\alpha}$$

If $H_a := \sigma_1^2 > \sigma_2^2$, then we define the rejection region as,

$$F > F_{\alpha}$$

And lastly, if $H_a := \sigma_1^2 \neq \sigma_2^2$, then we define the rejection region as,

$$F < F_{1-\alpha/2} \text{ or } F > F_{\alpha/2}$$